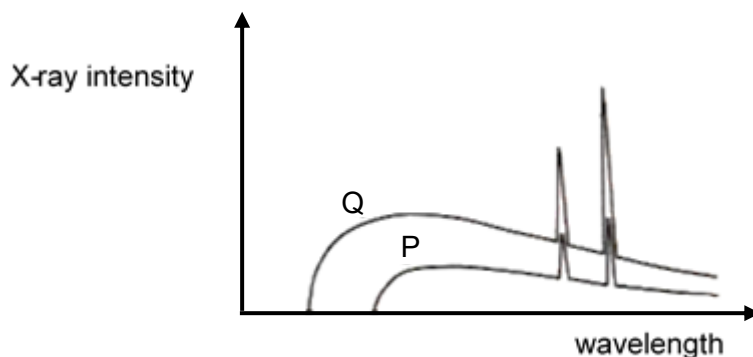


30. The given graph show the variation of X-ray intensity with wavelength emitted from two X-ray tubes P and Q.



Which of the following could be a deduction from the graphs?

- A** X-ray tube Q has the higher voltage applied to it and the target material in both tubes is the same.
- B** X-ray tube Q has the higher voltage applied to it and the target material in both tubes are different.
- C** X-ray tube P has the higher voltage applied to it and the target material in both tubes is the same.
- D** X-ray tube P has the higher voltage applied to it and the target material in both tubes are different.

Answer: A

The characteristic x-ray is the same (spikes are at the same wavelengths) so target material must be the same.

Q has a smaller minimum wavelength than P. Minimum wavelength when all k.e. of incident electron is converted to a single photon, so

$$e \square \text{ accelerating p.d.} = \frac{hc}{\lambda_{\min}}$$

$$\lambda_{\min} = \frac{hc}{e \square \text{ accelerating p.d.}}$$

Therefore, accelerating p.d. of Q is higher than P.